AudioLLM-D: Fine-tuning Audio LLM for Dementia Speech

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Abstract

Dementia speech contains disfluencies such as pauses, fillers, and repetitions that serve as important biomarkers of cognitive decline. However, general-purpose speech models often suppress these cues. We fine-tune Qwen2-Audio-7B-Instruct on DementiaBank using LoRA and a disfluency-aware vocabulary to preserve and transcribe dementia-specific speech patterns. We evaluate the model in two stages: (1) disfluency-preserving transcription and (2) audio question answering. Our best model achieves a WER of 0.334 and a disfluency F1 of 0.572, outperforming prompted baselines. While improved transcription alone does not yield strong diagnostic reasoning performance, the results demonstrate that preserving fine-grained acoustic biomarkers is a necessary step toward dementia-aware Audio LLMs.

1 Introduction

Dementia affects over 57 million people worldwide, yet most cases remain undiagnosed [14]. Speech is a promising non-invasive biomarker of cognitive decline, as people with dementia (PwD) often exhibit characteristic disfluencies such as pauses, fillers, repetitions, and fragmented sentences. However, modern ASR systems are typically trained to suppress these irregularities, removing information that may be clinically relevant.

This creates a dual challenge: voice-based models might fail to interpret the intent of PwDs, cutting them off from modern AI tools. Second, models smooth over the speech patterns that constitute essential evidence for reasoning about dementia diagnosis. Prior work such as WhisperD [1] and CrisperWhisper [18] improves transcription fidelity, but these systems are limited to speech recognition and cannot reason about audio content.

We address this gap by fine-tuning Qwen2-Audio-7B-Instruct [3] on DementiaBank with a disfluency-aware vocabulary and parameter-efficient adaptation. We evaluate the resulting model on both transcription and audio question-answering tasks. Our contributions are: (i) a disfluency-aware Audio LLM for dementia speech, (ii) an ablation study of training strategies for disfluency recovery, and (iii) an audio QA benchmark for evaluating reasoning over dementia speech.

2 Approach

Our model builds on Qwen2-Audio-7B-Instruct [3], which pairs a Whisper-large-v3 audio encoder [15] with a Qwen2 7B language model (LM). We extend the tokenizer with a dedicated set of disfluency tokens spanning different types of pauses, fillers, unclear speech, and non-verbal events, initialized from their string representations and kept trainable.

We fine-tune the model with parameter-efficient fine-tuning [12]. LoRA adapters are applied to the LM and to the audio-encoder layers, the audio-to-language projection is fully fine-tuned, and the input embeddings and LM head for the new disfluency tokens are kept trainable. Because disfluency tokens are rare, their training loss is upweighted so that the model is not biased toward dropping them. The dataset, disfluency vocabulary, and exact training configurations are detailed in the Experiments section 3.3.

We evaluate along two axes. First, transcription quality on raw audio against reference transcripts, compared against Whisper [16] and CrisperWhisper [18]. We wanted to benchmark against WhisperD [1] but it was not publicly available and the authors never responded to our requests. Second, diagnostic reasoning, assessed via QA pairs generated by an LLM from the reference transcripts.

3 Experiments

3.1 Dataset

We constructed our dataset from several sources within DementiaBank repository [2]. Preliminary analysis of the legacy Pitt Corpus revealed sig-

Token	Type	Train (%)	Test (%)
[uh]	filler	3.47	2.37
[unclear-speech]	intelligibility	2.91	1.91
[short-pause]	pause	1.09	1.52
[um]	filler	1.26	1.14
[long-pause]	pause	0.72	1.39
[medium-pause]	pause	0.44	0.77
[laughter]	non-verbal	0.29	0.58
[non-verbal]	non-verbal	0.27	0.35
Total		10.45	10.02

Table 1: Disfluency tokens and their share of all reference tokens in the training (32,461 tokens, 658 clips) and test (6,913 tokens, 168 clips).

nificant audio quality limitations, so we used the ADReSS challenge subset [10] with cleaner and denoised audio clips. We combined this with the Lu [9], Hopkins [6], DePaul [4], and Kempler [7] corpora, yielding 826 paired samples across 201 participants (≈5.45 hours of audio).

We developed a custom CHAT [11] parsing pipeline to isolate participant speech and map acoustic markers (pauses, fillers, unclear speech, non-verbal events) to eight dedicated tokens added to the Qwen2-Audio vocabulary (Table 1). To satisfy the encoder's 30-second input constraint, files were chunked by concatenating sequential utterances up to the threshold without truncation. An 80/20 speaker-disjoint train/test split was enforced to eliminate data leakage.

For the secondary evaluation, we assessed the usefulness of the transcripts on the base model's audio comprehension and reasoning via an audio-based Question Answering framework. QA pairs were systematically generated by Claude 3.5 Sonnet from ground truth transcripts, comprising two categories: content-based questions (multiple-choice and yes/no) testing literal comprehension, and disfluency-focused queries (count-based and yes/no) probing awareness of speech irregularities. Under TalkBank Ground Rules,[1] we report only aggregate statistics and model outputs.

3.2 Evaluation Method

To evaluate transcription quality, we use Word Error Rate (WER) alongside two disfluency F1 formulations. The first is a bag-of-tokens F1, which assesses whether the correct disfluency types appear in the transcript regardless of position. The second is an alignment-based order-aware (OA) F1, where the generated transcript is dynamically aligned to the reference before computing F1 over the resulting sequence, capturing both token identity and sequential accuracy. We compare against Whisper [16] and CrisperWhisper [18] on WER only, as neither system emits our disfluency vocabulary. QA performance is measured by accuracy, defined as the proportion of correctly answered questions.

[1] https://talkbank.org/0share/index.html

3.3 Experiment Details

We conduct a progressive series of experiments, each adding a single modification, to isolate the design choices that most influence disfluency recovery.

Zero-shot and few-shot. Before any fine-tuning, we benchmarked Qwen2-Audio-7B-Instruct model under zero-shot prompting (the model received only a system prompt defining the eight-token disfluency vocabulary), and few-shot prompting, (the same prompt supplemented with four hand-crafted transcription exemplars) and scored with WER and disfluency-token precision, recall, and F1. Both recovered very few disfluencies, establishing the need for fine-tuning.

Plain LoRA baseline. LoRA adapters (rank 16) are applied to all encoder and LM layers, with the newly introduced disfluency tokens kept trainable. The model is trained for 10 epochs with a learning rate of 8×10^{-5}, using a single shared learning rate across all trainable components and a constant disfluency token upweighting factor of 2.0 in the loss. This configuration represents the simplest disfluency-aware adaptation and serves as the primary baseline for all subsequent comparisons.

Token weighting and data augmentation. We replace the uniform token upweighting with a frequency-aware reweighting scheme, where each disfluency class is weighted inversely proportional to its empirical occurrence in the training set. Building on the weighted baseline, we introduce on-the-fly acoustic augmentation applied on the training data. Each training clip is perturbed with probability 0.5 using one of white noise, babble noise, reverberation, speed perturbation, or codec compression, sampled uniformly at random.

Synthetic data. We further augment training with synthetically generated, disfluency-rich speech to mitigate the scarcity of rare disfluency markers in the real dataset. Using ChatterBox TTS [17], we generate approximately one hour of synthetic data, where each utterance contains a single

disfluency type. This controlled setup allows the model to first learn each marker in isolation before encountering more naturalistic co-occurrences.

Focal loss on rare disfluencies. On top of the synthetic-data configuration, we introduce a frequency-tiered focal loss [8] ($\gamma = 2$) applied to the four rarest disfluency classes ([long-pause], [medium-pause], [laughter], [non-verbal]). This focuses learning on hard, under-recalled tokens by increasing gradient contribution from infrequent categories.

CrisperWhisper encoder. We additionally evaluate replacing the standard Whisper-large-v3 encoder with the CrisperWhisper encoder [18], which is fine-tuned for verbatim transcription of fillers and pauses. The hypothesis is that a verbatim-optimized encoder may provide a stronger acoustic foundation for disfluency modeling than the default noise-suppressing encoder.

MFA and CTC. We experimented with two auxiliary supervision signals; Montreal Forced Aligner (MFA) [13] and Connectionist Temporal Classification (CTC) [5]. First, we trained an auxiliary disfluency classifier using MFA time-aligned labels and injected its outputs into encoder representations to bias attention toward disfluency-relevant regions. Second, we added a CTC head to encourage frame-level alignment between acoustic features and disfluency tokens. But neither yielded consistent gains and hence were excluded from the final model.

Ablation Studies To identify which components are most critical for disfluency recovery, we conduct a set of ablation studies over LoRA adaptation scope. We compare with two configurations: (i) adapting only the language model backbone, (ii) adapting the language model backbone together with the multimodal projection layer.

3.4 Audio Question Answering

We evaluate the base model on the audio QA benchmark (Section 3.1). Each of the 672 questions (168 count, 168 multiple-choice, 336 yes/no) was answered under three conditions: audio-only (no transcript provided), audio augmented with ground-truth transcripts, and audio augmented with transcripts generated by our best fine-tuned model.

4 Results and Analysis

Table 2 reports results on both the training and test sets for the experimental progression described in the previous section. Several trends emerge.

4.1 Transcription

First, even the simplest disfluency-aware adaptation substantially outperforms the baseline systems. Plain LoRA achieves a WER of 0.37, improving over both Whisper's 0.51 and CrisperWhisper's 0.47, while simultaneously achieving strong disfluency detection performance (F1 of 0.51). In contrast, zero-shot and few-shot prompting of Qwen2-Audio produces negligible disfluency F1 scores, indicating that explicit fine-tuning is essential for recovering disfluency markers.

Starting with LM-only LoRA gives WER of 0.46 and F1 of 0.47. Extending adaptation to the multimodal projector gives WER of 0.49 and F1 of 0.52. Full finetuning (plain LoRA) improves WER to 0.37 with a comparable F1.

The largest improvement in disfluency detection comes from the combination of token re-weighting and acoustic data augmentation. Relative to the plain LoRA baseline, this configuration increases F1 from 0.51 to 0.55, demonstrating that both class imbalance and limited acoustic diversity are major bottlenecks for disfluency modeling.

MFA and CTC did not improve results. MFA guidance slightly reduced F1 despite lower WER, likely due to noisy forced-alignments and interference from feedback into the encoder. CTC also showed no consistent gains and was therefore not pursued further.

Subsequent modifications yield small but consistent gains. Adding synthetic disfluency-rich speech improves performance, with the best disfluency F1 of 0.5721 obtained using CrisperWhisper with synthetic data, while focal loss with the standard Whisper encoder achieves nearly identical F1 (0.5717) but the lowest WER (0.33). CrisperWhisper and focal loss therefore provide largely interchangeable benefits: each matches the other's F1 when paired with synthetic data, and their combination does not yield further improvements, indicating overlapping effects.

All models exhibit strong overfitting: the best system reaches 0.94 train F1 but only 0.57 on held-out speakers, a gap consistent across configurations. Despite improvements from data enrichment (reweighting, augmentation, synthetic

	Train			Test		
Model	**WER↓**	**F1↑**	**OA-F1↑**	**WER↓**	**F1↑**	**OA-F1↑**
External ASR / prompted baselines						
Whisper-large-v3	0.4601	–	–	0.5141	–	–
CrisperWhisper	0.4221	–	–	0.4665	–	–
Qwen2-Audio (zero-shot)	0.5656	0.0261	0.0071	0.5396	0.0331	0.0120
Qwen2-Audio (few-shot)	0.5899	0.0321	0.0104	0.6634	0.0289	0.0100
Our fine-tuning progression						
LLM-only LoRA	0.0838	0.8748	0.8546	0.4649	0.4750	0.3063
LLM LoRA + Multimodal projector	0.0381	0.9384	0.9323	0.4902	0.5196	0.3318
Plain LoRA	0.0048	0.9895	0.9876	0.3692	0.5097	0.3754
CTC objective + Different learning rates	0.0507	0.9088	0.8937	0.4710	0.4959	0.3193
Data Augmentation & Weighted Loss	0.0212	0.9406	0.9351	0.3803	0.5523	0.3921
MFA classifier	0.0437	0.9396	0.9340	0.3653	0.5133	0.3513
Synthetic Data + CrisperWhisper Encoder	0.0410	0.9251	0.9167	0.3566	**0.5721**	0.3989
Synthetic Data + CrisperWhisper + Focal Loss	0.0199	0.9473	0.9427	0.3619	0.5531	0.3956
Synthetic Data + Base Encoder + Focal Loss	0.0169	0.9403	0.9356	**0.3340**	0.5717	**0.4216**

Table 2: Train and Test Results

Condition	**Overall↑**	**Yes/No↑**	**Multiple Choice↑**	**Count↑**	**Content↑**	**Disfluency↑**
Audio-only	0.5714	0.6845	0.6964	0.2202	0.7887	0.3542
Audio + Gold transcript	0.5729	0.7381	0.7202	0.0952	0.7708	0.3750
Audio + Fine-tuned transcript	0.5372	0.6786	0.7024	0.0893	0.7411	0.3333

Table 3: Audio QA accuracy

speech), gains remain marginal, suggesting that limited labeled dementia speech is the primary bottleneck.

4.2 Question-Answer task

Table 3 reports audio QA accuracy across three inference conditions. Content-oriented questions (yes/no and multiple-choice) are answered reasonably well under all conditions, approaching 0.74–0.79 accuracy. Count questions, which require precise memory of disfluency events, are substantially harder (0.22 audio-only). Adding gold transcripts yields only marginal overall gains (0.571 $\rightarrow$ 0.573) and actually degrades count accuracy (0.22 $\rightarrow$ 0.095), likely because the model conflates disfluency tokens in context with ordinary words. Fine-tuned transcripts perform worse still (0.537), consistent with error propagation from transcription imperfections. These results indicate that transcription improvements alone have not bridged the gap to reliable diagnostic reasoning, and that a jointly optimized transcription and QA objective may be needed.

5 Ethical considerations

This work aims to improve accessibility for people with dementia (PwD), whose speech is often poorly handled by conventional voice AI. DementiaBank data was used under TalkBank's guidelines. The primary risk is misuse for diagnosis or surveillance, therefore, the model should be considered a research prototype requiring clinician oversight and informed consent. Additionally, the model is English-only and trained on a limited demographic, restricting its broader applicability.

6 Conclusion

We presented a disfluency-aware Audio LLM for dementia speech by fine-tuning Qwen2-Audio-7B-Instruct on DementiaBank. Our best model achieved a WER of 0.334, disfluency F1 of 0.572, and order-aware F1 of 0.422, outperforming prompted and external ASR baselines in recovering clinically relevant speech irregularities. Ablations show that encoder adaptation, data augmentation, synthetic disfluency-rich speech, and focal loss all improve disfluency recovery, while MFA- and CTC-based auxiliary objectives do not. Although gains in transcription quality do not yet translate to strong audio reasoning performance, preserving disfluencies is an important step toward dementia-aware Audio LLMs. Future work should prioritize collecting or curating additional dementia speech data to improve generalization as the primary bottleneck identified in our analysis is data scarcity.

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A Appendix

A.1 Token-Level Analysis

The model learns frequent disfluency markers well, with fillers such as `[um]` and `[uh]` achieving the strongest performance on the held-out set. In contrast, pause markers, `[unclear-speech]`, `[laughter]`, and `[non-verbal]` remain substantially more challenging.

The largest source of error is under-prediction: the model often produces a cleaner transcript than the reference by omitting pauses and non-verbal events when the surrounding utterance remains intelligible. When the model does detect pauses, their duration e.g., `[long-pause]` predicted as `[medium-pause]`, suggesting the model is better at recognizing the presence of a pause than its exact category.

We also observe two characteristic failure modes for `[unclear-speech]`. The model occasionally replaces unclear regions with plausible lexical content, while in other cases it uses `[unclear-speech]` as a fallback when uncertain about a word. Rare events such as `[laughter]` and `[non-verbal]` are frequently omitted, indicating that their acoustic signatures remain difficult to learn from limited training data. Finally, errors often compound in long or highly disfluent utterances: once the lexical transcript diverges from the reference, subsequent disfluency markers become misaligned, leading to multiple token-level errors within the same example.

Model	uh	um	short pause	medium pause	long pause	unclear speech	laughter	non verbal
	164	79	105	53	96	132	40	24
LLM-only LoRA	0.70	0.77	0.42	0.31	0.41	0.33	0.16	0.29
LLM + projector	0.70	0.74	0.45	0.25	0.39	0.45	0.32	0.29
Plain LoRA	0.76	0.75	0.42	0.26	0.56	0.32	0.29	0.42
CTC objective	0.67	0.78	0.48	0.30	0.43	0.37	0.19	0.31
Data Aug. + Weighted Loss	0.75	0.76	0.45	0.32	0.48	0.42	0.39	0.43
MFA classifier	0.71	0.75	0.33	0.29	0.44	0.43	0.36	0.35
Synth. + CrisperWhisper	0.76	0.80	0.43	0.43	0.51	0.47	0.44	0.29
Synth. + Crisper + Focal	0.75	0.79	0.36	0.22	0.46	0.48	0.47	0.39
Synth. + Base + Focal	0.75	0.76	0.43	0.35	0.48	0.51	0.44	0.41

Table 4: Per-token count-based F1 on the test set

Model	uh	um	short pause	medium pause	long pause	unclear speech	laughter	non verbal
	164	79	105	53	96	132	40	24
LLM-only LoRA	0.54	0.71	0.20	0.17	0.17	0.09	0.00	0.21
LLM + projector	0.57	0.71	0.18	0.13	0.15	0.11	0.07	0.06
Plain LoRA	0.64	0.73	0.21	0.16	0.38	0.12	0.00	0.28
CTC objective	0.52	0.74	0.25	0.18	0.19	0.08	0.09	0.11
Data Aug. + Weighted Loss	0.62	0.74	0.17	0.14	0.32	0.17	0.28	0.22
MFA classifier	0.58	0.72	0.16	0.16	0.23	0.14	0.09	0.11
Synth. + CrisperWhisper	0.61	0.76	0.21	0.24	0.35	0.18	0.07	0.10
Synth. + Crisper + Focal	0.60	0.76	0.20	0.20	0.31	0.15	0.15	0.29
Synth. + Base + Focal	0.64	0.73	0.21	0.16	0.34	0.23	0.15	0.38

Table 5: Per-token order-aware F1 on the test set

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